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Industry-First Structural Enhancement for Noise-Optimized Seismic Resolution Imaging

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Abstract

This study introduces a novel technique to enhancing seismic image resolution by applying various embossing filter techniques to an edge-detected seismic dataset, created by pre-processing the seismic volume using an edge detection algorithm to highlight critical geological features such as faults, fractures, and stratigraphic discontinuities. To effectively enhance seismic reflection resolution, we employ four distinct embossing filter techniques: kernel scaling, smoothing composition, Fractional Embossing Using Frequency-Domain Directional Derivatives (fractional directional embossing) and Amplitude Optimization Technique. By leveraging the unique strengths of each technique, we can suppress noise and enhance signal in different domains, ultimately leading to a more accurate and detailed seismic image. Our results demonstrate the efficacy of each technique, highlighting their respective strengths and weaknesses providing valuable insights for the oil and gas industry and informing the development of optimized seismic data processing workflows.

Introduction

Background

Seismic imaging plays a vital role in hydrocarbon exploration, providing interpreters with 2D and 3D views of subsurface structures that are essential for accurate resource evaluation. The quality of these interpretations, however, depends on two key factors: the quality of seismic data and the effectiveness of post-processing and image enhancement techniques. In this context, techniques that enhance the visibility of subtle structural features, such as faults, horizons, and stratigraphic features, are particularly valuable. One such technique is the embossing filter, a directional enhancement method that uses convolution to emphasize intensity gradients in seismic data, thereby improving the delineation of these critical features. By leveraging the embossing filter's capabilities, interpreters can gain a more detailed understanding of subsurface structures, ultimately informing more accurate and effective exploration and production strategies.

Mathematically, the convolution operation can be represented as:

$$(f * g)(x, y) = \sum \sum f(x - m, y - n)g(m, n)$$

where:

- $f(x, y)$ is the input image
- $g(x, y)$ is the filter kernel
- (x, y) are the coordinates of the output pixel
- (m, n) are the coordinates of the filter kernel
- $\sum\sum$ represents the summation over all possible values of m and n

The embossing filter kernel is typically a small matrix that is designed to emphasize intensity gradients along a specific direction. For example, a simple (3×3) embossing kernel might look like this:

$$K = \begin{bmatrix} -2 & -1 & 0 \\ -1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

This kernel emphasizes intensity gradients along the top-left to bottom-right direction, making it useful for enhancing faults and fractures in seismic data.

The embossing filter, while effective in enhancing seismic features, can also amplify noise, making it challenging to distinguish between meaningful structural features and unwanted noise. To address this limitation, optimizing the embossing filter to reduce noise amplification is crucial. The complexity of seismic imaging, as noted by [Yilmaz \(2001\)](#), requires careful consideration of multiple factors, including data acquisition, processing, and interpretation. Furthermore, the use of embossing filters in seismic data processing has been extensively discussed in the literature, with authors such as [Gonzalez and Woods \(2008\)](#) providing comprehensive overviews of digital image processing techniques. Building on this foundation, our study focuses on enhancing the emboss filter to improve the interpretability of seismic data. To achieve this goal, we investigate the application of several techniques, including kernel scaling, smoothing composition, fractional directional embossing, and amplitude optimization technique

By applying these techniques to an edge-detected seismic dataset, we aim to evaluate their effectiveness in reducing noise and enhancing seismic data interpretability, ultimately contributing to a more comprehensive understanding of subsurface structures. The results of this study will provide valuable insights into the strengths and weaknesses of each technique, informing the development of optimized seismic data processing workflows and improving subsurface interpretation accuracy.

Problem Statement

Embossing filtering is a powerful image enhancement technique capable of emphasizing structure by accentuating directional gradients in seismic data ([Gonzalez and Woods, 2008](#)). However, its application to seismic data presents significant challenges due to the inherent complexity of seismic images. The filter operates by enhancing contrasts, which can lead to the indiscriminate uplift of both meaningful features and high-frequency noise. This noise amplification reduces the interpretability of the data and can mislead interpreters by masking subsurface geological features, such as faults, fractures, and stratigraphic boundaries.

Therefore, a clear need for a methodology that can:

- Focus enhancement efforts on structurally relevant features, such as faults, fractures, and stratigraphic boundaries, while suppressing noise and other irrelevant features.
- Avoid or minimize the uplift of noise, which can reduce the accuracy of interpretations and lead to incorrect decisions.

To address these challenges, this study explores the application of different techniques to optimize the emboss filter, including:

- Kernel scaling: a technique that involves scaling the filter kernel to adjust the level of enhancement and reduce noise amplification.
- Smoothing composition: a technique that involves combining the emboss filter with a smoothing filter to reduce noise and improve the overall quality of the image.
- Fractional directional embossing: a technique that involves applying the emboss filter in a fractional manner to enhance specific features or directions.
- Amplitude optimization technique: applies a tailored adjustment to the coherence data to mitigate the noise and artifacts generated by the embossing process

By applying these techniques separately to an edge- detected seismic dataset (also known as coherence), we aim to evaluate their effectiveness in reducing noise and enhancing the interpretability of seismic data. The ultimate goal of this research is to develop a robust and reliable methodology for emboss filtering that can improve the accuracy and reliability of seismic interpretations, leading to better decision-making and more accurate reservoir characterization. The industry current workflow when using embossing filter is shown below:

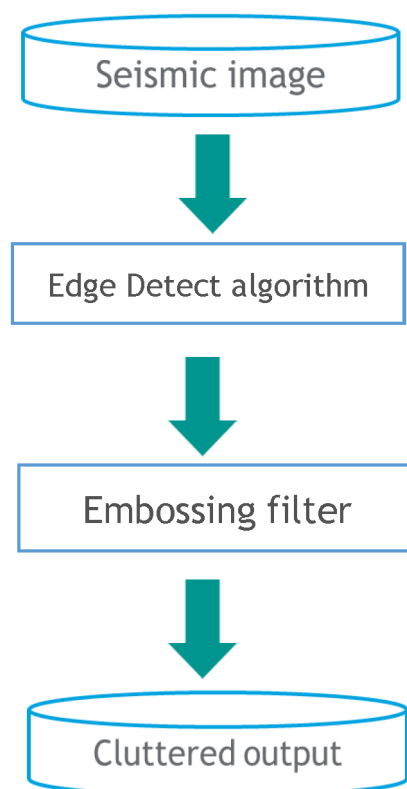


Figure 1—industry current workflow

Objective

The primary objective of this study is to develop and validate a noise-optimized seismic enhancement workflow that maximizes structural clarity while minimizing noise amplification. Our framework leverages a structurally focused seismic volume generated using edge detection filtering, which highlights critical seismic faults, fractures, and stratigraphic discontinuities.

We apply four different techniques separately to edge-detected dataset to evaluate their effectiveness in enhancing seismic attributes and reducing noise: kernel scaling, Composited filtering, Fractional directional embossing. We evaluate each technique based on its ability to improve the signal-to-noise ratio, enhance subtle features, and maintain consistency across slices. Through a comparative analysis, we identify the optimal technique for achieving clean, sharp, and interpretable seismic volumes that are ready for structural and stratigraphic interpretation, and ultimately, for reservoir characterization and petrophysical analysis.

Methodology

The enhancement framework described in this work involves four key techniques, each applied to a seismic volume that had already undergone edge detection preprocessing. This preprocessing step utilizes edge detection-based filtering to isolate structural contrasts, such as faults, reflectors, and stratigraphic boundaries, while suppressing amplitude variations. By doing so, each subsequent technique can operate more selectively on meaningful features.

Dataset

The dataset used in this study is edge-detected version from the original seismic dataset through edge detection-based algorithm (it is also known as coherence). the edge-detected dataset used as input similar to the one described by Yi Luo et al. (2013) in their paper "Generalized Hilbert transform and its applications in geophysics". This preprocessing step was crucial for guiding the subsequent structural filtering techniques described in the following sections. It ensured that enhancement operations were focused on geologically significant contrasts while avoiding unnecessary amplification of amplitude-based noise.

Coherence attributes are widely recognized as effective edge detection methods in 3D seismic interpretation, particularly for fault and channel mapping (Marfurt et al., 1998). They function by quantifying the similarity between neighboring seismic traces, where low coherence indicates discontinuities, such as faults or stratigraphic terminations.

The coherence volume is calculated by computing the localized waveform similarity along inline and crossline directions, producing a 3D cube of coherence coefficients (Luo et al., 1996). These coefficients can be interpreted similarly to seismic amplitudes, with lineaments of low coherence highlighting fault surfaces even when no fault reflection is visible. Advanced coherence techniques, such as eigenvalue-based techniques (Gersztenkorn and Marfurt, 1996) and dip estimation (Marfurt and Kirlin, 2000), have further improved the ability to detect subtle structural features

Most of the techniques tested using a time slice from the Norway dataset, in [figure 2a](#) is shown the raw original time slice. While in [figure 2b](#) you can see the detected image. In [figure 2c](#) is the time slice after applying the embossing filter, you can see in the red cycle how noisy it is:

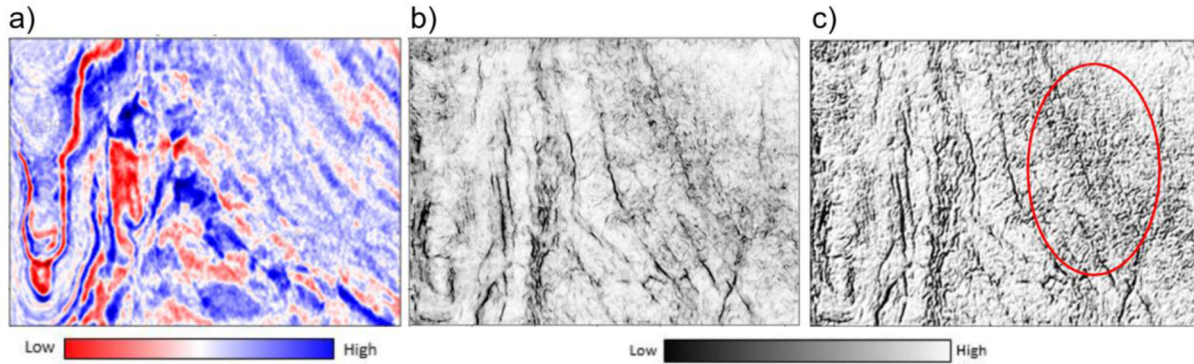


Figure 2—Chosen time slice through (a) raw seismic amplitude, (b) edge detection algorithm, (c) embossing filter

Kernel Scaling

To address the noise uplift problem inherent in traditional, emboss filtering, the first enhancement technique involved adjusting the strength of the emboss operation itself. This was done by applying a uniform scaling factor to all elements of the emboss kernel. The aim was to maintain directional structure enhancement while suppressing the intensity of noise amplification.

Background. The emboss filter is a directional enhancement operator commonly applied via 2D convolution. A typical emboss kernel emphasizes intensity gradients along a diagonal or linear direction. For example:

$$K = \begin{bmatrix} -2 & -1 & 0 \\ -1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

This kernel highlights contrasts in the top-left to bottom-right direction and, when applied to seismic slices, reveals subtle structural trends. However, the filter's non-selective gradient amplification causes unwanted uplift of high-frequency noise—especially in regions of low signal strength.

Method 1. To reduce this side effect, a scalar multiplier x was applied uniformly to the kernel:

$$K' = x \cdot K$$

Where:

- K' is the scaled kernel,
- $x \in \mathbb{R}$ is a scalar between 0 and 1,
- K is the original emboss kernel.

A range of values were tested for the scalar multiplier x , from 0.1 to 0.9, with increments of 0.1. This allowed for a thorough evaluation of the effect of kernel scaling on noise reduction and feature enhancement. Two specific values, $x = 0.3$ and $x = 0.5$, were selected for detailed analysis.

Results and Observations. Visual inspection of the scaled emboss kernels with $x = 0.3$ and $x = 0.5$ revealed distinct differences in their performance. Although the difference between the two was not drastic, the kernel with $x = 0.3$ produced a noticeable blurry effect on the noise, while simultaneously uplifting the

features, as illustrated in Figure 3a. In contrast, the kernel with $x = 0.5$ resulted in slightly more pronounced feature uplift, accompanied by a moderate reduction in noise, as depicted in Figure 3b. Notably, both scaled kernels demonstrated significant improvement over the original emboss kernel, which was prone to noise amplification, as shown in Figure 2c. These results suggest that scaling the emboss kernel can be an effective means of reducing noise and enhancing feature visibility, but the optimal scaling factor may depend on the specific application and desired outcome.

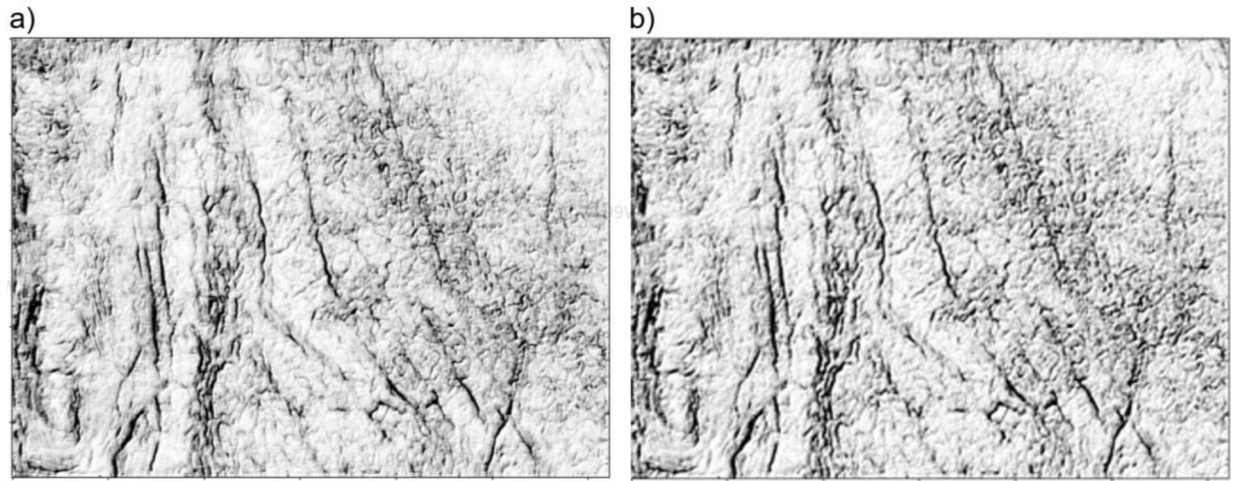


Figure 3—comparison between level of soothing when (a) $x = 0.3$ and when (b) $x = 0.5$

Composed Filtering

Given the limitations of kernel scaling, the second technique involved combining the emboss filter with a smoothing operator to dampen noise before edge enhancement. This technique, known as composed filtering, aims to reduce the noise amplification associated with traditional emboss filtering while preserving the structural definition of the seismic image. Although kernel scaling provided better results, the composed filtering technique was still explored to investigate its potential in improving the emboss filter, as it offers a different technique to noise reduction and feature enhancement.

Background. Composed filtering involves convolving the emboss kernel with a smoothing kernel to form a new composed kernel. The smoothing kernel is designed to suppress local variations and reduce noise, while the emboss kernel is used to enhance the structural definition of the seismic image. By combining these two kernels, the composed filtering technique can produce a more balanced result, with reduced noise and preserved structural definition. According to Liu et al. (2013), composed filtering can be an effective method for reducing noise and enhancing features in seismic images. They demonstrated that the use of a smoothing kernel can help to reduce the noise amplification associated with traditional emboss filtering, while preserving the structural definition of the image.

The design of the smoothing kernel is critical to the success of the composed filtering technique. A Gaussian kernel is often used as the smoothing kernel, as it is effective at reducing noise and preserving features. The standard deviation of the Gaussian kernel can be adjusted to control the amount of smoothing applied to the image. A smaller standard deviation will result in less smoothing, while a larger standard deviation will result in more smoothing.

k =

-2	-1	0
-1	1	1
0	1	2

Method 2. The emboss kernel k was defined as:

Two different smoothing kernels were used to compose with the emboss kernel:

G1: Gaussian-like Smoothing Kernel G1 is a 3x3 Gaussian-like smoothing kernel, which is a type of kernel commonly used in image processing for smoothing and blurring images. The kernel is defined as:

G1 =

0.125	0.25	0.125
0.25	0.5	0.25
0.125	0.25	0.125

This kernel has a Gaussian-like distribution, with the center pixel having the highest weight (0.5) and the surrounding pixels having decreasing weights. The kernel is symmetric, meaning that it has the same weights for pixels on either side of the center pixel. G1 is a relatively mild smoothing kernel, meaning that it will not excessively blur the image, but will still help to reduce noise and smooth out small-scale features.

G2: Gaussian-like Smoothing Kernel with Reduced Weights G2 is also a 3x3 Gaussian-like smoothing kernel, but with reduced weights compared to G1. The kernel is defined as:

G2 =

0.0625	0.125	0.0625
0.125	0.25	0.125
0.0625	0.125	0.0625

This kernel has a similar Gaussian-like distribution to G1, but with reduced weights for all pixels. The center pixel still has the highest weight (0.25), but it is lower than the corresponding weight in G1 (0.5).

The design of the smoothing kernels was based on the work of [Gonzalez and Woods \(2008\)](#), who demonstrated the effectiveness of Gaussian-like smoothing kernels in image processing. The use of two different smoothing kernels, G1 and G2, allowed for a comparison of the effects of different smoothing strengths on the composed kernel. The smoothing kernels G1 and G2 were convolved with the emboss kernel K to form a new composed kernel:

$$K_{composed}(m, n) = \sum G(i, j) \cdot E(m - i, n - j)$$

The resulting composed kernels are:

K_composed1 =

- 0.125	-0.3125	-0.25	-0.0625	0
-0.3125	-0.6875	-0.375	0.0625	0.125
-0.25	-0.375	0	0.375	0.25
-0.0625	0.0625	0.375	0.8125	0.3125
0	0.125	0.25	0.3125	0.125

K_composed2 =

-0.222	-0.333	-0.333	0.111	0
-0.333	-0.333	-0.222	0.111	0.111
-0.333	-0.222	0.111	0.444	0.3333
-0.111	0.111	0.444	0.555	0.333
0	0.111	0.333	0.333	0.222

Results and Observations. A visual comparison between [Figure 4a](#), [Figure 4b](#), and the embossed image ([Figure 2c](#)) reveals that the results are almost indistinguishable from each other. Both [Figure 4a](#) and [Figure 4b](#) show minimal improvement over the original embossed image, with no noticeable reduction in noise or significant enhancement of structural features.

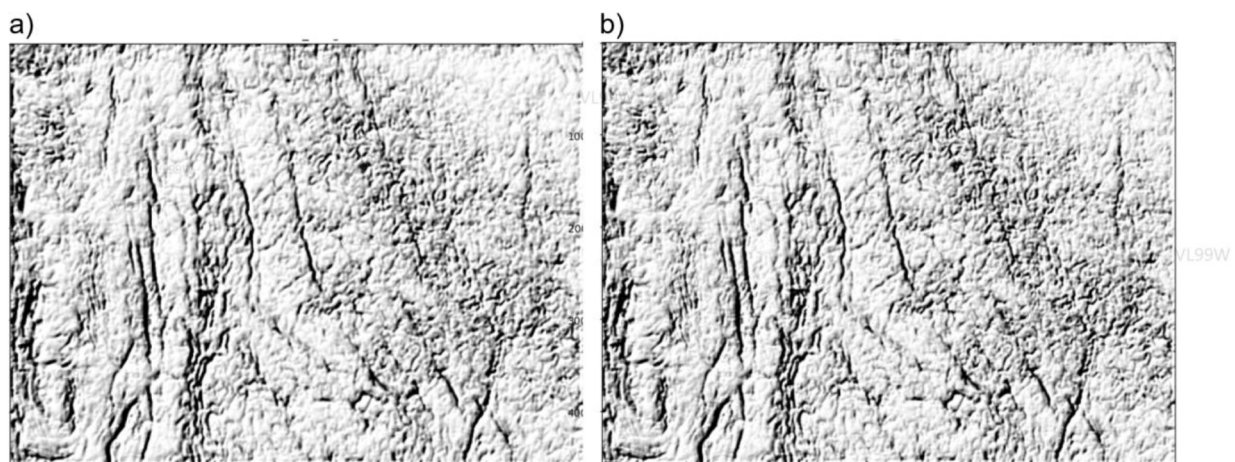


Figure 4—Comparison between level of soothing with kernel (a) K_composed1 and kernel (b) K_composed2

Furthermore, the results suggest that both G1 and G2 produce similar outcomes, with no discernible difference between the two. This suggests that the choice of smoothing kernel may not have a significant impact on the outcome of the composed filtering technique in this particular application. According to [Gonzalez and Woods \(2008\)](#), the choice of smoothing kernel can affect the outcome of image processing techniques, but in this case, the results suggest that the composed filtering technique is relatively robust to the choice of smoothing kernel.

Fractional Embossing Using Frequency-Domain Directional Derivatives

The third technique involved using fractional embossing with frequency-domain directional derivatives to enhance the seismic image. This technique allows for controlled directional sharpening and can be used to emphasize specific features in the image. The advantages of this technique include its ability to enhance specific features in the image while reducing noise, and its flexibility in terms of the direction and order of the derivative.

Background. Fractional derivatives have gained significant attention in recent years due to their ability to model complex systems and signals with non-integer order dynamics. In this study, we present a numerical method for computing fractional derivative operators, which can be utilized in various seismic signal processing applications. The proposed technique uses a frequency-domain formulation to calculate the fractional derivative operator, allowing for efficient computation and flexibility in choosing the derivative order. We use fractional derivative implemented by Al-Dossary and Marfurt in 2006.

A fractional derivative of order α in the x-direction is represented in the Fourier domain as:

$$dp/dx \approx F - \mathbb{I}[ikx F(p)]$$

Where:

- p is the seismic attribute
- kx is the wavenumber in the x-direction
- F is the Fourier transform operator
- F^{-1} is the inverse Fourier transform operator
- α is the power of the fractional derivative

This operator enhances gradients aligned with the x-direction, allowing for controlled directional sharpening. The use of fractional derivatives in seismic attribute analysis can provide improved results in certain applications, such as edge detection and feature enhancement.

Alternatively, the fractional derivative can be approximated by a 1st derivative followed by a low-pass filter:

$$dap/dxa \approx F - \mathbb{I}[i(kx)aF(p)] * LPF$$

Where LPF is a low-pass filter.

This estimate can be used to control the direction of the fractional derivative, allowing for more accurate and effective sharpening of the seismic attributes.

Method 3. The implementation workflow for fractional embossing using frequency-domain directional derivatives involves the following steps:

- Fractional Derivative Operator: Compute the fractional derivative operator
- Application of the Fractional Derivative Operator: Apply the fractional derivative operator to the Fourier transform of the image to compute the fractional derivative of the image.
- Inverse Fourier Transform: Compute the inverse Fourier transform of the result to transform it back into the spatial domain.
- Embossing: Apply a standard emboss kernel to the normalized image to further enhance the features.

The parameter used in this implementation is alpha which is the fractional order of the derivative, which controls the amount of sharpening applied to the image. The output of this implementation is an embossed image that highlights the features of the original image in the specified direction.

Results and Observations. The fractional embossing technique's outcomes are presented in Figures 5a-f, which show six images corresponding to α values ranging from 0.1 to 1.2. Increasing α values progressively elevate the background noise and features, effectively leveling the ground of the time slice. This results in a diminished contrast between features and background, indicating that the technique may not be effective for enhancing features or suppressing noise in seismic images, as it amplifies background noise and features rather than selectively enhancing features of interest.

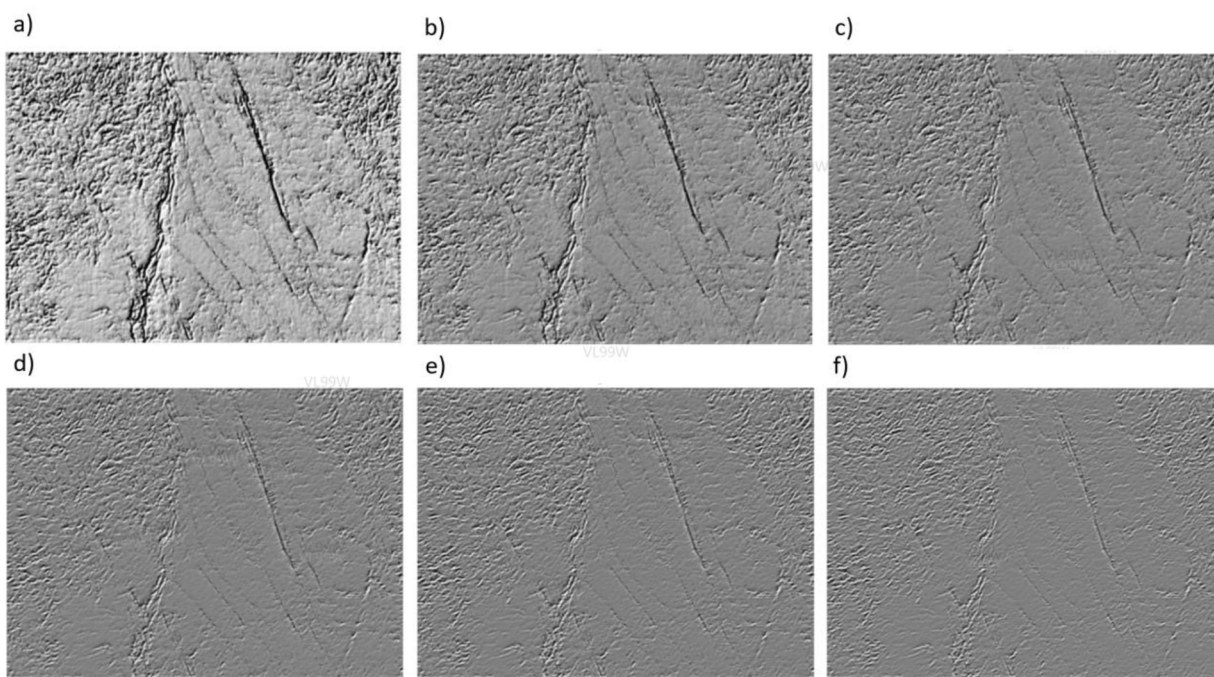


Figure 5—Results of fractional embossing 5a-f for α values from 0.1 to 1.2

Amplitude Optimization Technique

The fourth and final method involves a specialized amplitude optimization technique, which is applied as a post-processing step to the embossing filter. This technique is designed to mitigate the noise and artifacts generated by the embossing process, while preserving the essential characteristics of the input data. By refining the spectral characteristics of the data and adapting to the local properties of the image, this technique aims to improve the interpretability of seismic images.

Background. The amplitude optimization technique is a post-processing method that builds upon the output of an embossing filter, which enhances the visibility of subsurface features in seismic images. The technique involves a non-linear operation that adjusts the amplitude of the embossing filter output to better reveal subtle features and reduce noise. This process can be seen as a form of enhanced contrast enhancement, where the goal is to improve the visibility of features of interest while reducing the impact of noise and artifacts. The application of this technique in seismic image processing is designed to enhance the visibility of faults, fractures, and stratigraphic variations, ultimately leading to more accurate interpretations and better-informed decision-making.

Method 4. The amplitude optimization technique involves a unique transformation that refines the spectral characteristics of the data. This transformation is applied to the output of the embossing filter, and is designed

to improve the visibility of coherence areas and result in a more balanced visualization of the subsurface. The technique may involve multi-scale processing, anisotropic filtering, and non-linear diffusion, which can be used to capture features at different scales and resolutions, account for the directional nature of seismic data, and reduce noise while preserving features. By integrating these various components, the amplitude optimization technique provides a powerful tool for improving the quality and interpretability of seismic images.

Results and Observations. The application of the amplitude optimization technique to the seismic data resulted in a significant improvement in the quality and interpretability of the images. The technique was able to effectively mitigate the noise and artifacts generated by the embossing process, while preserving the essential characteristics of the input data as shown in [figure 6](#). The resulting images showed a marked improvement in the visibility of faults, fractures, and stratigraphic variations, making it easier to interpret the subsurface geology. The technique also demonstrated a high degree of robustness and flexibility, making it suitable for a wide range of seismic imaging applications. Overall, the amplitude optimization technique proved to be the most effective method for improving the quality and interpretability of seismic images, and is a valuable tool for geoscientists and engineers working in the field of seismic imaging.

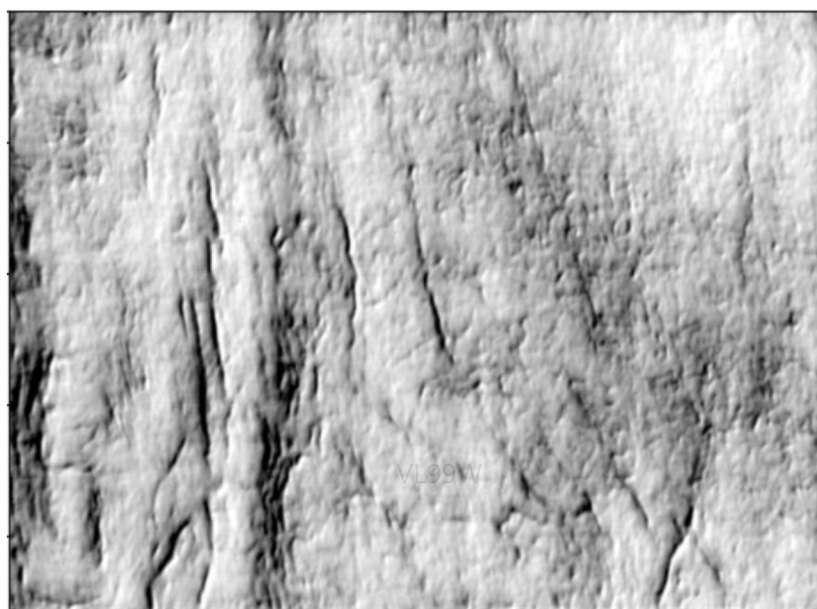


Figure 6—the image after optimizing the amplitude then applying emboss filter

Co-rendering in seismic data interpretation is a valuable visualization technique that enables the simultaneous display of multiple seismic attributes—such as amplitude, coherence, and acoustic impedance—within a single integrated view. By leveraging distinct color maps, transparency controls, or blending strategies, co-rendering allows interpreters to observe structural and stratigraphic features in greater context. This integrated visualization facilitates more efficient interpretation by eliminating the need to switch between separate volumes, thereby reducing cognitive load and interpretation time. Moreover, it enhances the visibility of subtle subsurface features that may be overlooked when attributes are examined individually. Ultimately, co-rendering supports a more comprehensive and accurate understanding of the subsurface, improving confidence in geological and reservoir characterization.

Below in [figure 7a-b](#) is comparison of the Co-rendering result with the amplitude optimization and without:

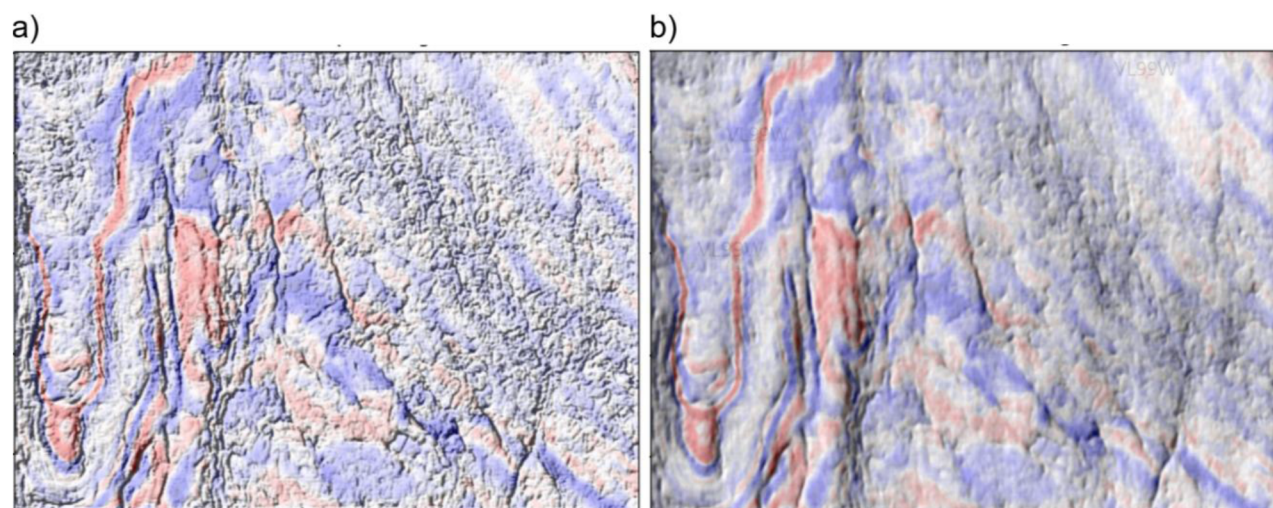


Figure 7—co-rendering image with raw seismic image embossing filter (a) and optimizing amplitude technique (b)

Results

The proposed workflow for seismic image enhancement using the amplitude optimization technique is as follows:

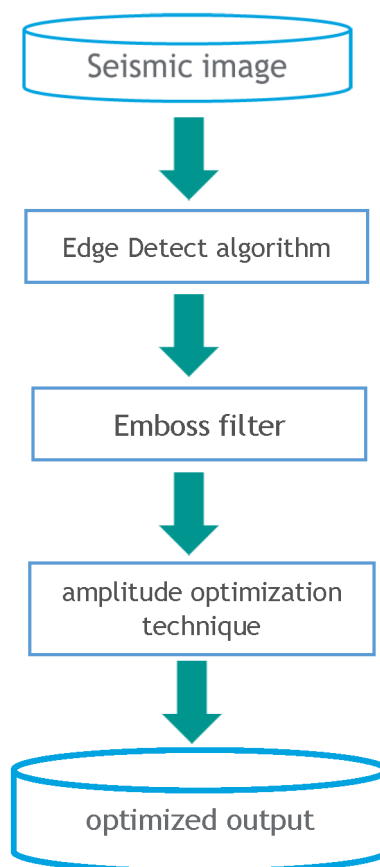


Figure 8—our proposed workflow.

The addition of the amplitude optimization step significantly improves the quality and interpretability of the seismic images, making it a valuable tool for geoscientists and engineers working in the field of seismic imaging.

The four different 3D volume is shown in 9a-d below of the work tested on another dataset:

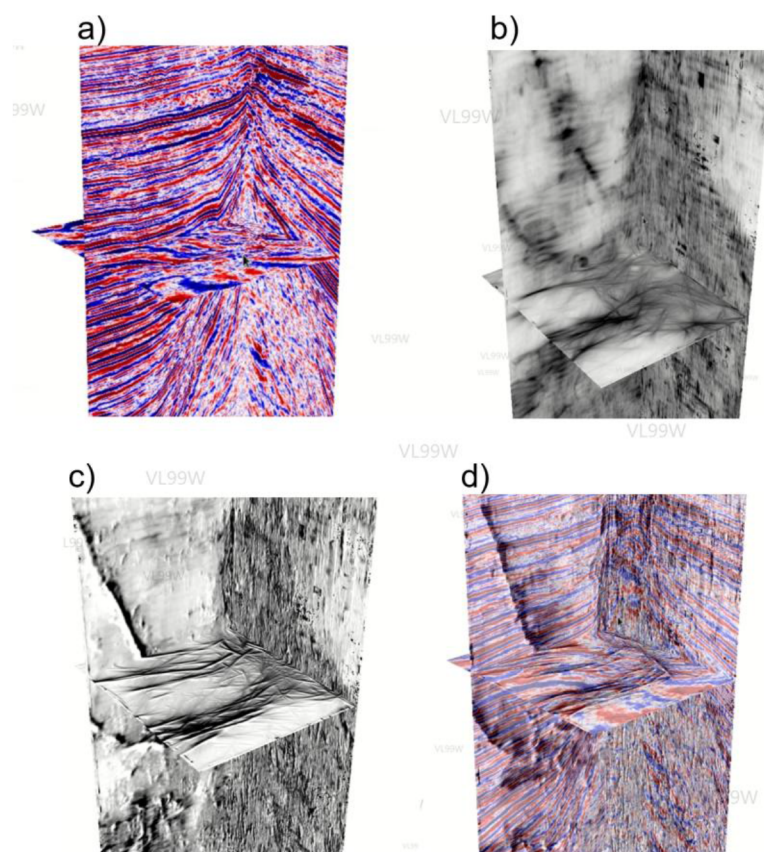


Figure 9—raw volume (a), edge detection algorithm applied in (b) and optimized amplitude embossing (c) co-rendering raw volume and volume optimized amplitude embossing

Conclusion

This study presents a comprehensive evaluation of four distinct techniques for enhancing the quality and interpretability of seismic images: kernel scaling, composed filtering, fractional embossing using frequency-domain directional derivatives, and amplitude optimization technique. The results of this investigation demonstrate that each technique exhibits unique strengths and weaknesses, and the selection of an optimal technique is contingent upon the specific characteristics of the seismic data and the desired outcome.

A comparative analysis of the first three technique reveals that kernel scaling emerges as the most effective technique among them, yielding a significant improvement in noise reduction and feature enhancement. In contrast, the composed filtering technique exhibits minimal improvement over the original embossed image, with negligible reduction in noise or enhancement of structural features. Furthermore, the fractional embossing technique is found to amplify background noise and features, rather than selectively enhancing features of interest, thereby limiting its efficacy in seismic image enhancement.

However, the amplitude optimization technique surpasses the other techniques in terms of its overall performance, resulting in a substantial enhancement of image quality and interpretability. This technique demonstrates a remarkable ability to mitigate noise and artifacts generated by the embossing process, while preserving the essential characteristics of the input data. The ensuing images exhibit a pronounced improvement in the visibility of faults, fractures, and stratigraphic variations, thereby facilitating a more accurate interpretation of the subsurface geology.

In conclusion, this study provides a rigorous evaluation of the efficacy of various techniques in enhancing seismic image quality when using embossing filter. The findings of this research highlight the

exceptional performance of the amplitude optimization technique, which emerges as a superior technique for improving the accuracy and reliability of seismic interpretations. The amplitude optimization technique offers a robust and flexible technique for a wide range of seismic imaging applications, and its application can have a profound impact on the oil and gas industry, enabling more informed decision-making and optimizing exploration and production strategies. Future research directions may focus on further refining and optimizing this technique, as well as exploring its applicability to diverse seismic data sets and imaging applications.

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